Differential Equations - Notes

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Part I
Basic Concepts

1 Definitions

1.1 What is a differential equation?

- A differential equation is an equation that contains derivatives, either ordinary or partial derivatives
- A popular example is newtons second law of motion, $F = ma$
- This is a differential equation since $a = \frac{dv}{dt}$
- Given this, we can actually write Newton's second law as a differential equation: $m\frac{dv}{dt} = F(t, v)$

1.2 What is Order?

- The order is the largest derivative present in the differential equation.
- For example, $\frac{\partial^3 u}{\partial x^2 \partial t} = 1 + \frac{\partial u}{\partial y}$ is a third order differential equation since it's greatest derivative is 3.

1.3 Ordinary vs. Partial Differential Equation

- An ordinary differential equation only has original differential equations, ode.
- A partial differential equation will have partial derivatives in it.

1.4 Linear Differential Equation

A linear differential equation is a differential equation following the form:

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)} + ... + a_1(t)y'(t) + a_0(t)y(t) = g(t)$$

1.5 Initial Conditions

An initial condition allows us to determine which solution we're after. For example, $y(t_0) = y_0$ or $y^{(k)}(t_0) = y_k$.

1.6 Implicit vs. Explicit Solutions

An implicit solution is a solution given in the form of $y = y(t)$
An explicit solution is a solution that isn't implicit.

Part II
First Order Differential Equations

2 Linear Differential Equations

2.1 Introduction

A linear first order differential equation is an equation in the form:

$$\frac{dy}{dt} + p(t)y = g(t)$$

- Here, we can assume $p(t)$ and $g(t)$ are continuous.

Now, let's assume the existence of a variable, $\mu(t)$, where we multiple all factors by it

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y = \mu(t)g(t)$$

To solve problems using this:

- Put the differential equation in the correct form above.
- Find the integrating factor, $\mu(t)$, where $\mu(t) = e^{\int p(t)dt}$
- Multiply everything in the differential equation with $\mu(t)$, and verify the leftside becomes the product rule, $(\mu(t)y(t))'$.
- Integrate both sides.
- Solve for $y(t)$.

2.2 Example 1

Let's try to find the solution to the following differential equation:

$$\frac{dv}{dt} = 9.8 - 0.196v$$

We can see that we need to put the equation into linear form.

$$\frac{dv}{dt} + 0.196v = 9.8$$

Let's solve for $\mu(t)$

$$\mu(t) = e^{\int 0.196 dt} = e^{0.196t}$$

Next, we can plug in $\mu(t)$ in all variables

$$e^{0.196t}\frac{dv}{dt} + e^{0.196t}0.196v = 9.8e^{0.196t}$$

We can rearrange some numbers:

$$e^{0.196t}\frac{dv}{dt} + 0.196e^{0.196t}v = 9.8e^{0.196t}$$

this resembles the product rule, hence we can say:

$$(e^{0.196t}v)' = 9.8e^{0.196t}$$

Now, we can integrate:

$$\int (e^{0.196t}v)'dt = \int 9.8e^{0.196t}dt$$

$$e^{0.196t}v = 9.8\int e^{0.196t}dt$$

for the right equation, we can use u sub

$$e^u dt, \text{ where } u = 0.196t,\ du = 0.196dt, \text{ and } dt = \frac{du}{0.196}$$

We can plug this into our equation

$$e^{0.196t}v + k = 9.8 \cdot \frac{1}{0.196}\int e^u du$$
$$e^{0.196t}v + k = 50\int e^u du$$
$$e^{0.196t}v + k = 50e^{0.196t} + c$$

Finally, we can solve for y, or v in this context:

$$v = \frac{50e^{0.196t} + c}{e^{0.196t}}$$

3 Separable Equations

3.1 Introduction

A seperable differential equation is an equation where you can separate the variables:

$$N(y)\frac{dy}{dx} = M(x)$$

To put it simply, all the ys need to be on one side, xs on the other. To solve such a differential equation, you'd integrate both sides.

$$\int N(y)\frac{dy}{dx}dx = \int M(x)dx$$

We can use substitution to get this instead:

$$\int N(u)du = \int M(x)dx$$

3.2 Example

Solve the following differential equation and determine the interval of validity for the solution.

$$\frac{dy}{dx} = 6y^2x, \text{ and } y(1) = \frac{1}{25}$$

The first step is seperating the Xs and Ys

$$dx \cdot \frac{dy}{dx} = 6y^2x \cdot dx$$
$$\frac{dy}{y^2} = \frac{6y^2x \cdot dx}{y^2}$$
$$\frac{1}{y^2}dy = 6xdx$$

From here, we can do integration:

$$\int \frac{1}{y^2}dy = \int 6xdx$$
$$\int y^{-2}dy = 3x^2 + C$$
$$-y^{-1} = 3x^2 + C$$
$$-\frac{1}{y} = 3x^2 + C$$

now, we can solve for y:

$$y = \frac{1}{3x^2+C}$$

Finally, we can solve for C to get our differential equation. This is trivial, as we have $y(1) = \frac{1}{25}$

$$\frac{1}{25} = \frac{1}{3+C}$$

We can easily figure out that $C = 22$.

Hence our final differential equation is:

$$y = \frac{1}{3x^2+22}$$

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